



DEVELOPMENT, TUNING AND VALIDATION OF A CLOSED-LOOP WHEEL BRAKE SLIP CONTROL SYSTEM

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2022UME1109

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Company Overview

Aeronautical Development Establishment



- Founded in 1959 by Govt. of India
- Research & Development in the field of military aviation

Technologies used for project:

- Digital Flight Computer
- Matlab/Simulink

Company details

- Tagline:
Balasya Mulam Vigyanam
(Knowledge is the Foundation of Strength)

- Headquartered in
Bengaluru, Karnataka

Major Products developed by ADE DRDO



DRDO Ghatak



DRDO Nishant

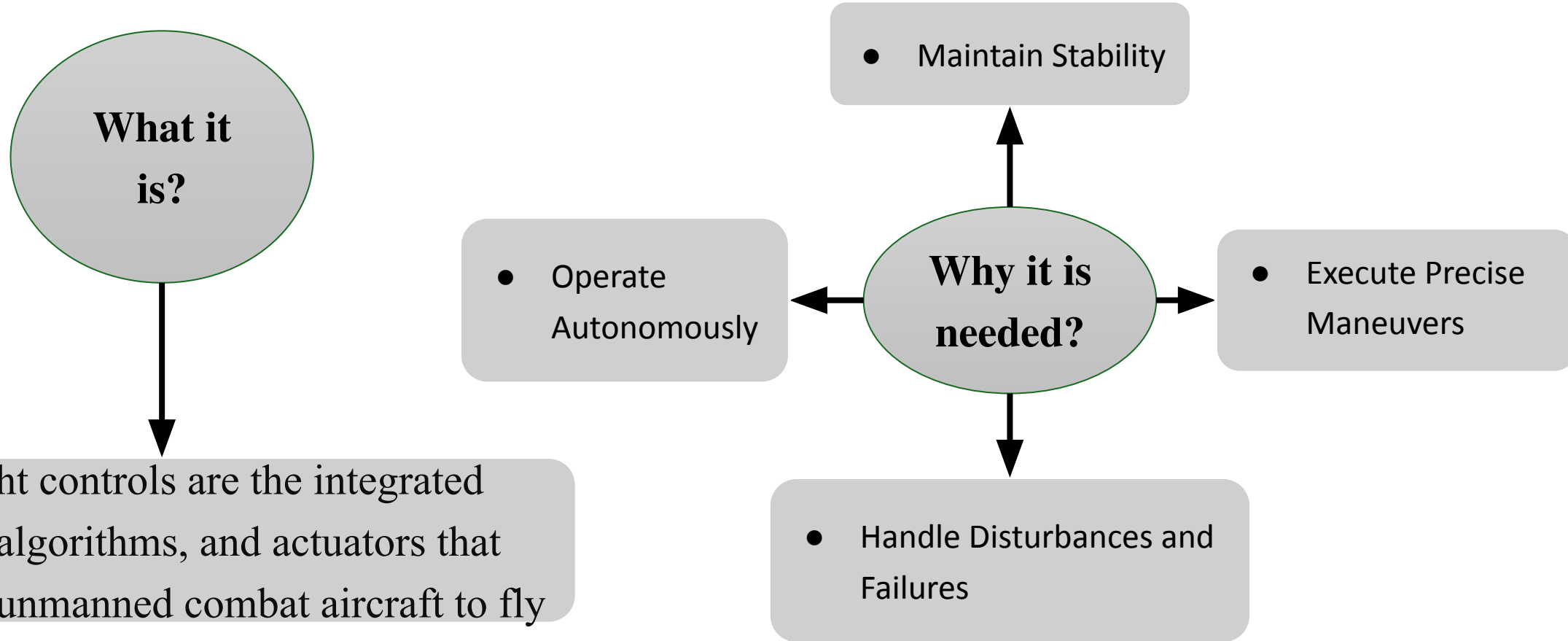


DRDO TAPAS

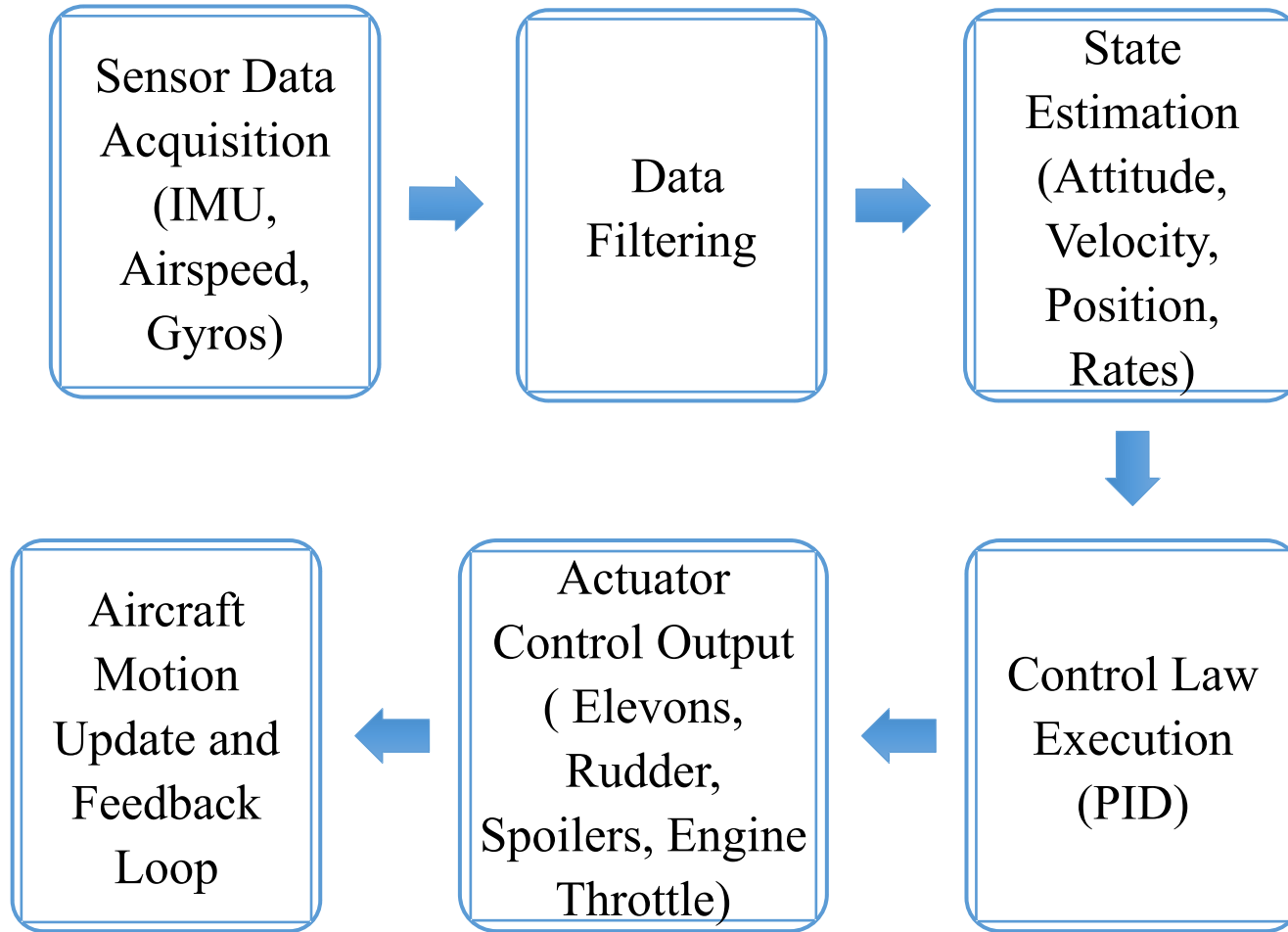


DRDO Lakshya

Meaning & Purpose of UAV Flight Controls



How Flight Control Computer Works?



Actuator controlled using FCC

Aircraft

- **Flight & motion:** Fixed-wing UAVs use rigid wings and forward motion for efficient flight with six degrees of freedom.
- **Actuators:** Ailerons, elevator, rudder, and throttle control flight, while flaps, air brakes, and landing gear support takeoff and landing.
- **Control system:** The flight control computer coordinates all actuators to ensure stability and mission execution.

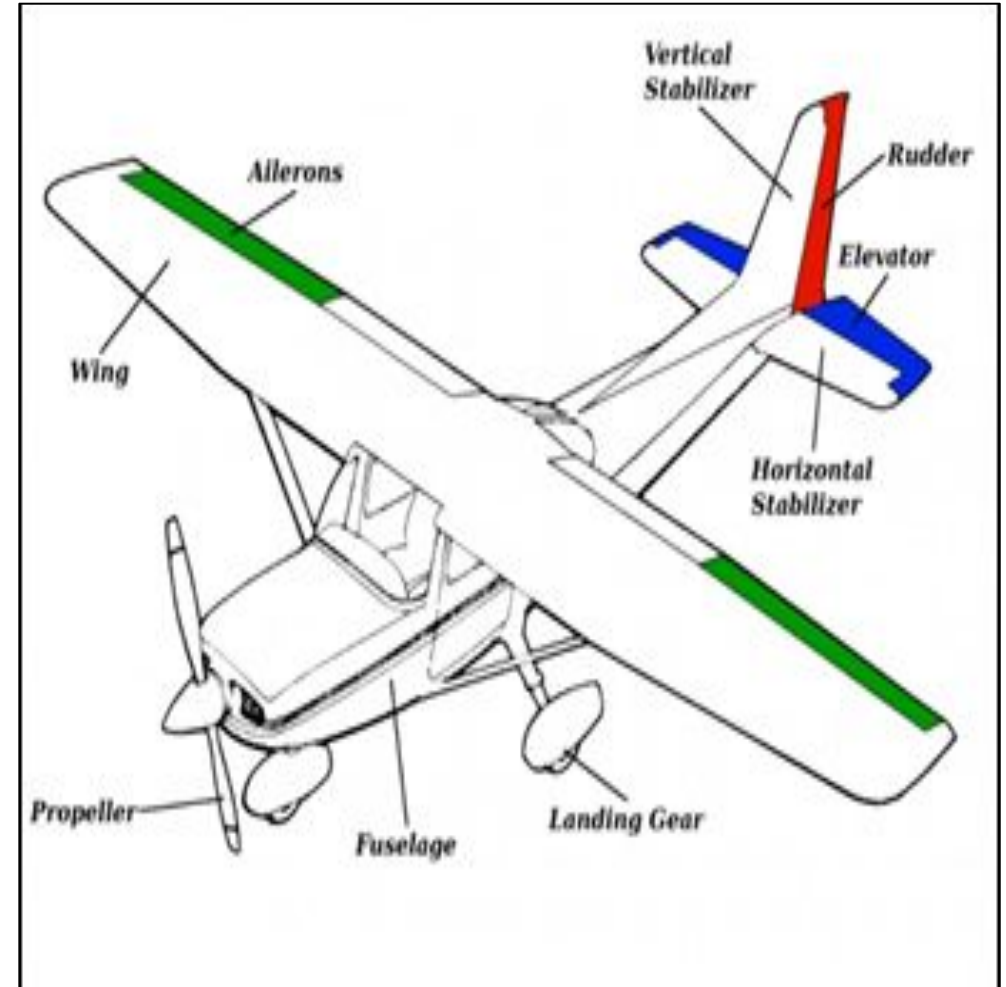


Fig: Fixed wing aircraft with actuators

Structural Components of Actuator Controls

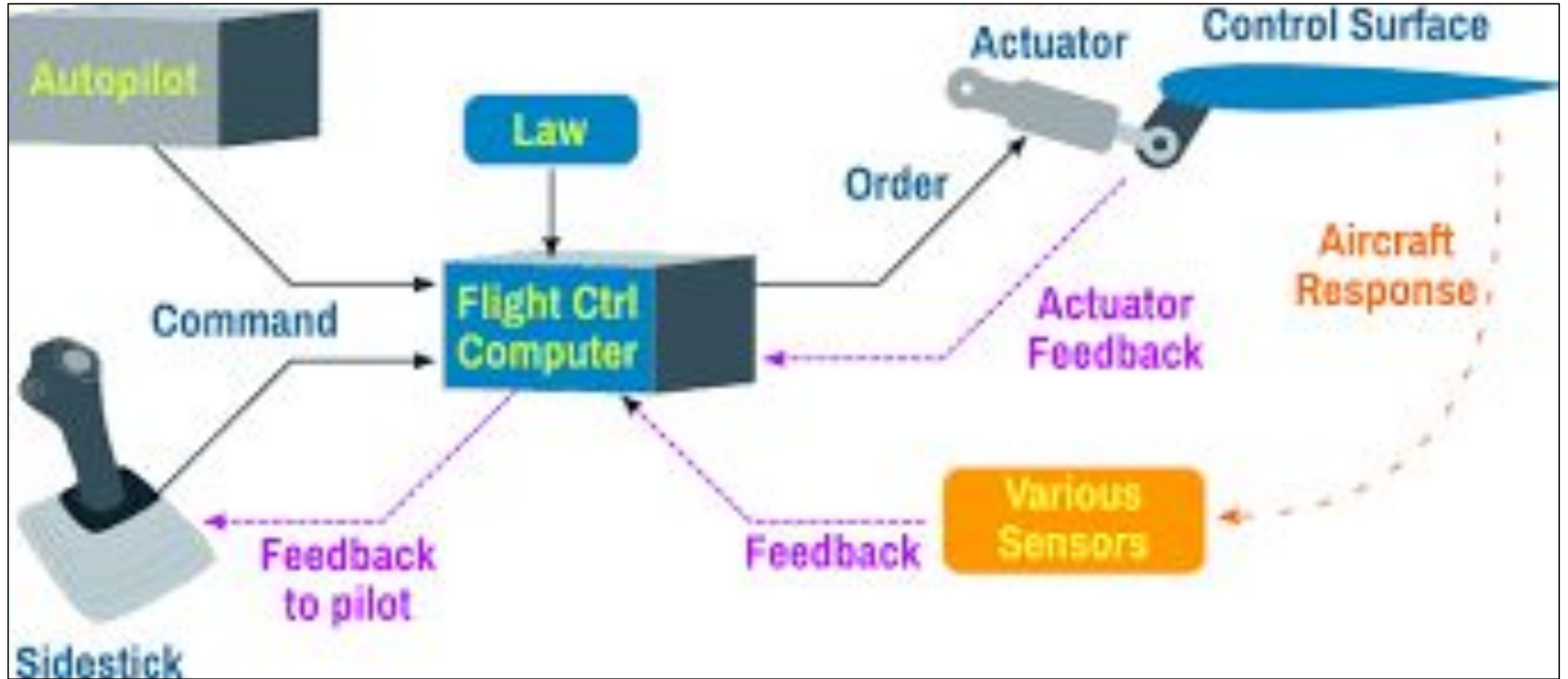


Fig: Components for unmanned air vehicle flight controls

Components of Braking

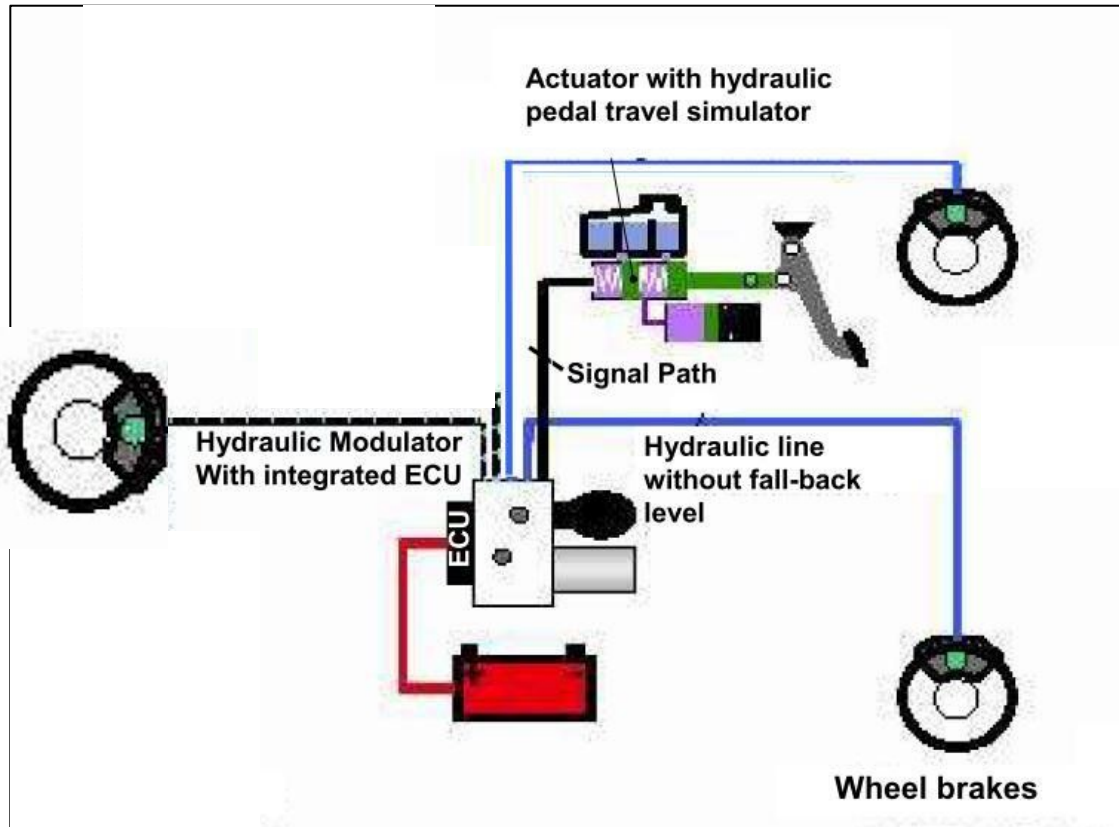


Fig: Functional Components of Braking

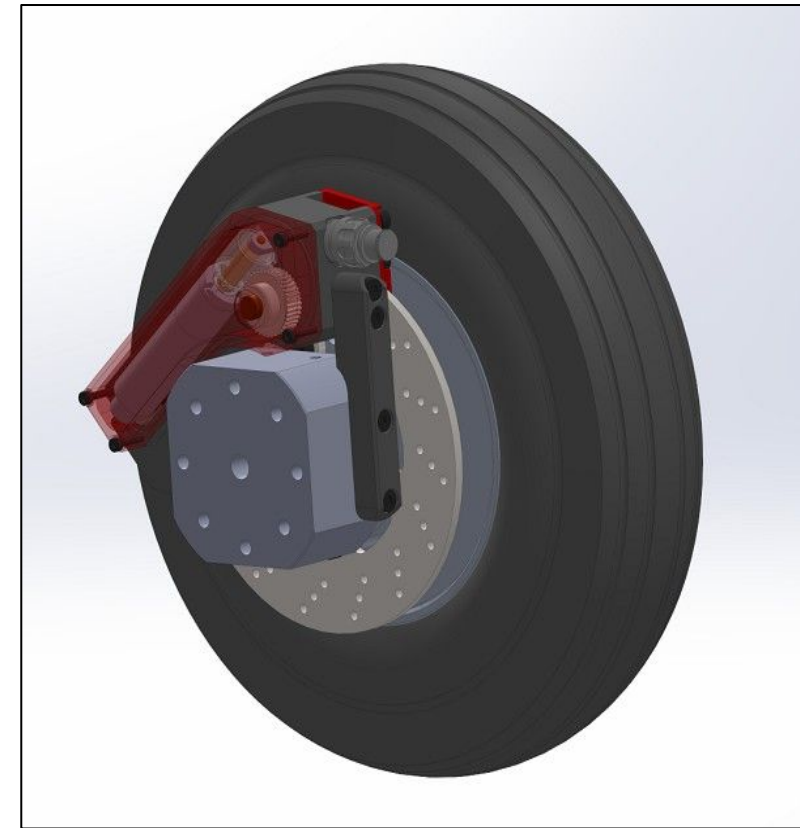


Fig: Physical Components of Brake

Develop, Tune and Validate a closed-loop wheel brake slip control system

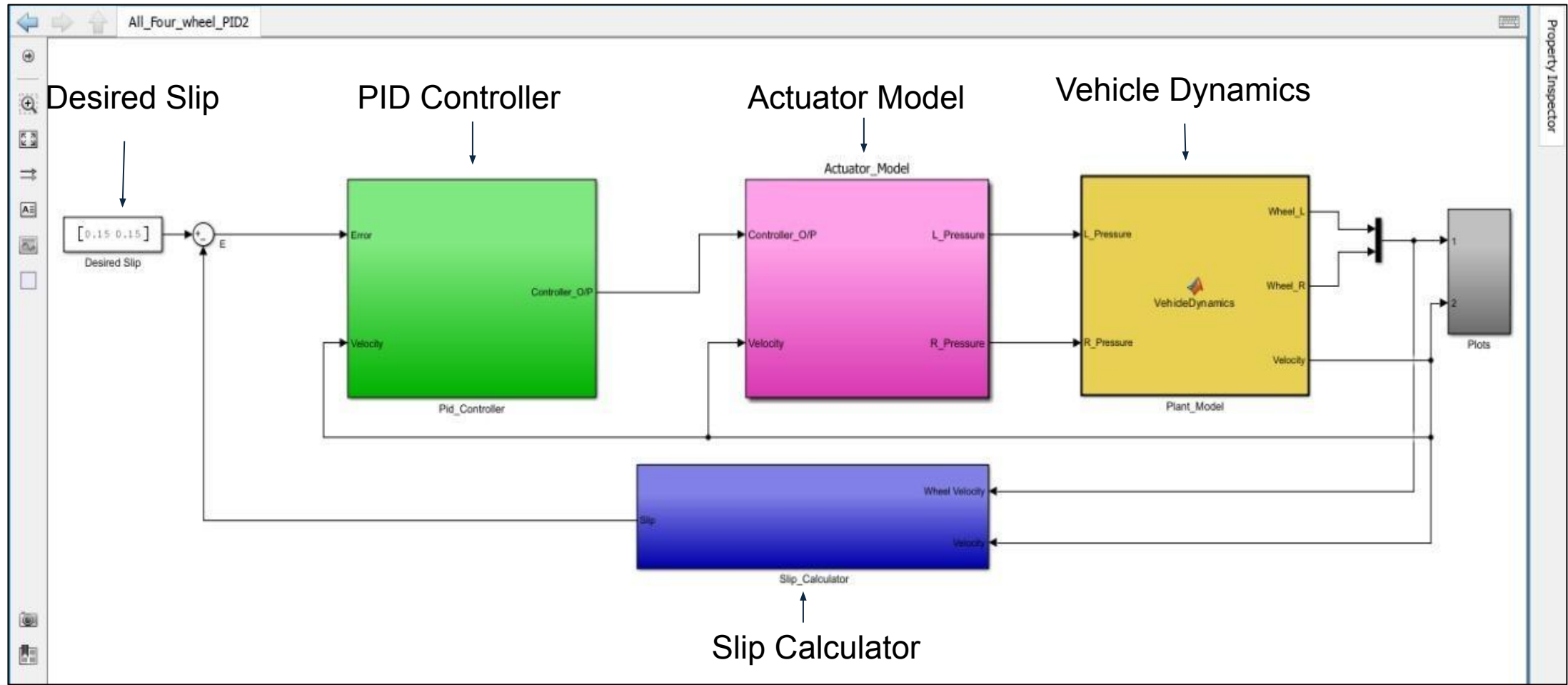


Fig: Modelled architecture for closed loop wheel brake slip control system on simulink

PID Controller Block

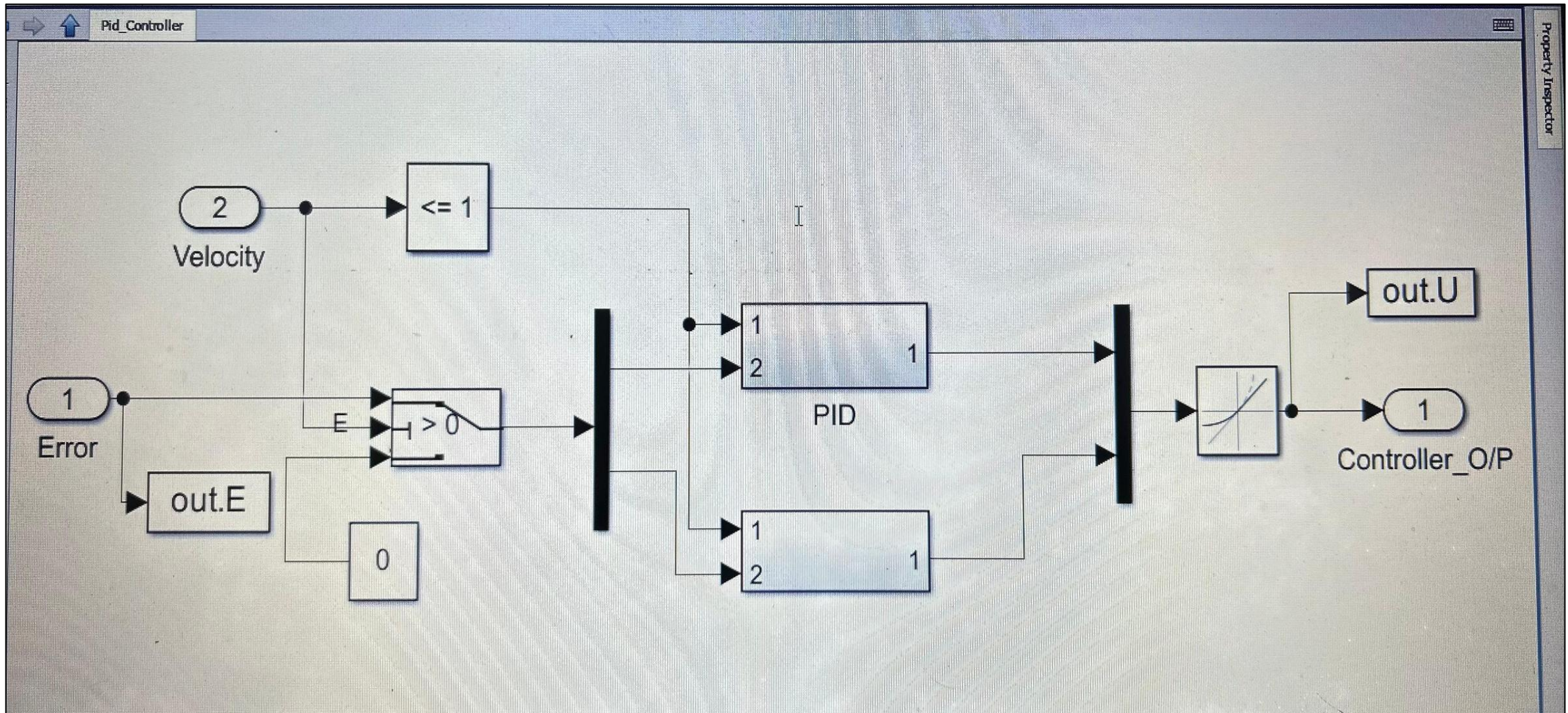


Fig: This block represents the internal logic used for building controller

Slip Calculation

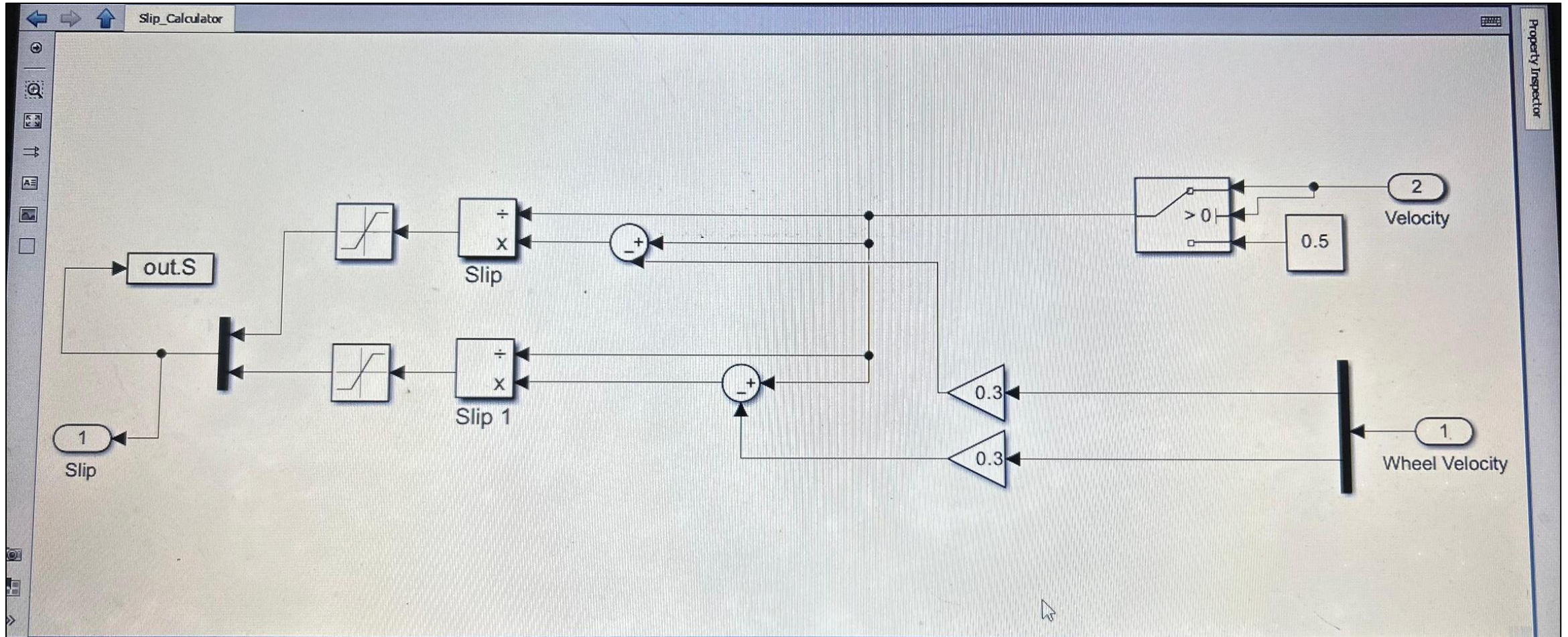


Fig: This block represents the modelled computation for calculation of slip

Actuator Dynamics Modelling

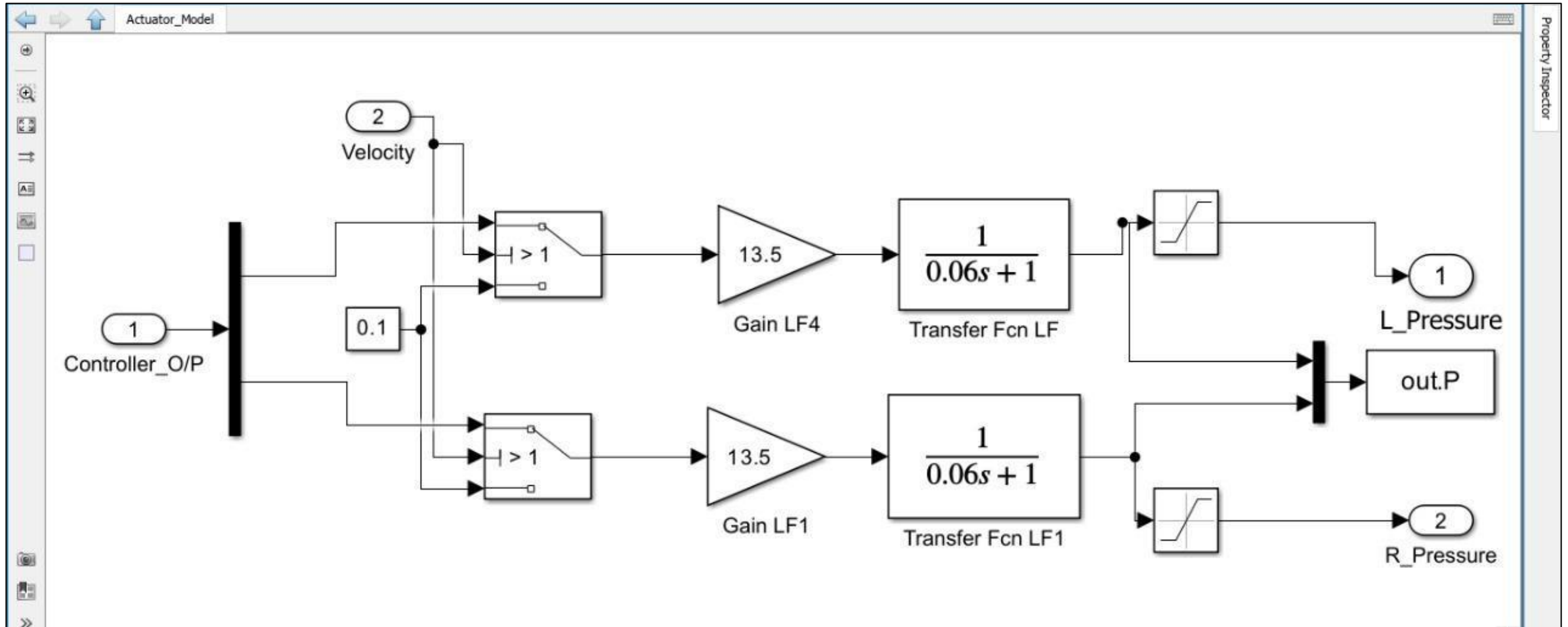


Fig: This block represents the physical modeling of the actuators used for braking

Plant Model

```
Plant_Model
function [Wheel_L, Wheel_R, Velocity] = VehicleDynamics(L_Pressure, R_Pressure)

% Vehicle parameters
r = 0.30;      % wheel radius (m)
m = 1500;      % UAV mass (kg)
I = 2.0;       % wheel inertia (kg.m^2)
Kb = 75;       % brake gain
b_w = 1.0;     % wheel damping
dt = 0.001;    % simulation time step (s)
g = 9.81;     % gravity (m/s^2)

Wheel_max = 100;
Wheel_min = 0;
Vel_max = 30;
Vel_min = 0;

% Initialization
persistent Wheel_L_prev Wheel_R_prev Velocity_prev
if isempty(Wheel_L_prev)

    Velocity_prev = 20;      %m/s
    Wheel_L_prev = Velocity_prev / r;    % rad/s
    Wheel_R_prev = Velocity_prev / r;    % rad/s

end
```

Fig: This block represents the vehicle dynamics used for study

Plant Model

```
27
28 % Brake torques
29 Tbr = [Kb*L_Pressure, Kb*R_Pressure];
30
31 % Wheel rotational dynamics
32 Wheel_prev = [Wheel_L_prev, Wheel_R_prev];
33 Wheel = Wheel_prev + ((-Tbr - b_w*Wheel_prev)/I)*dt;
34 Wheel = min(max(Wheel, Wheel_min), Wheel_max);
35
36 Wheel_L = Wheel(1);
37 Wheel_R = Wheel(2);
38
39 % Wheel slip calculation
40 slip_vec = max((Velocity_prev - Wheel*r) / max(Velocity_prev, 0.1), 0);
41 slip_vec = min(slip_vec, 1);
42
43 % Tyre friction coefficient
44 mu_vec = 1.0 - 0.8 * slip_vec;
45 mu_vec = max(mu_vec, 0);
46 mu_vec = min(mu_vec, 1);
47
48 % Longitudinal braking forces
49 NormalLoad = m * g / 2;
50 Fxf_vec = mu_vec .* NormalLoad;
```

Fig: This block represents the vehicle dynamics used for study

Plant Model

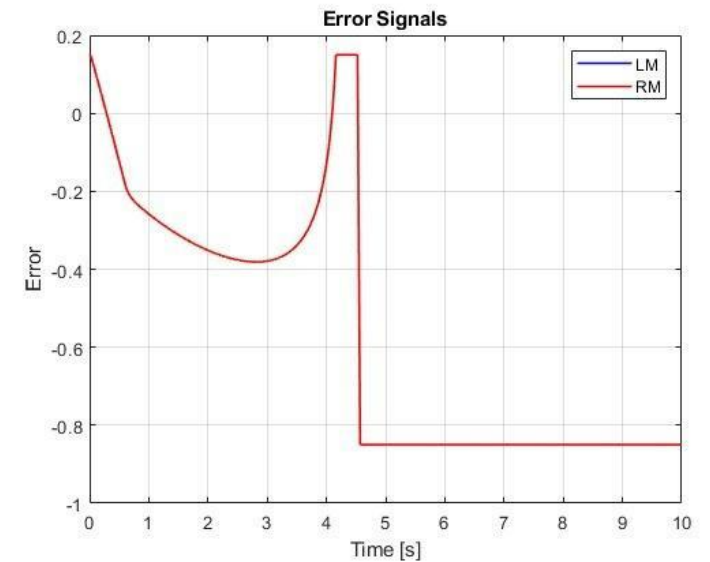
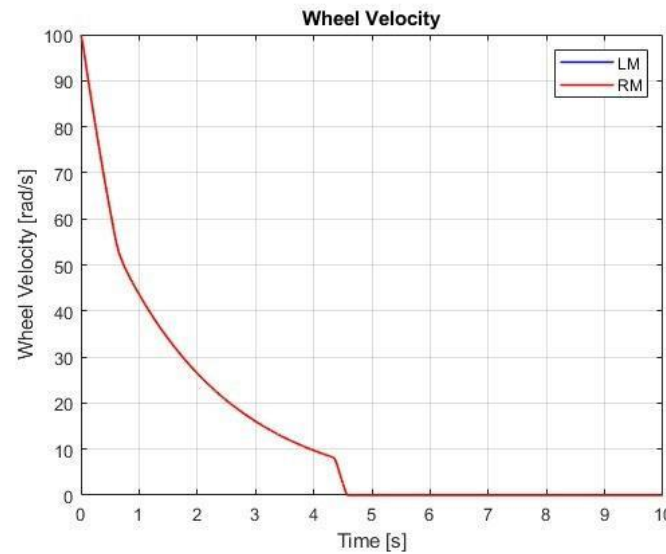
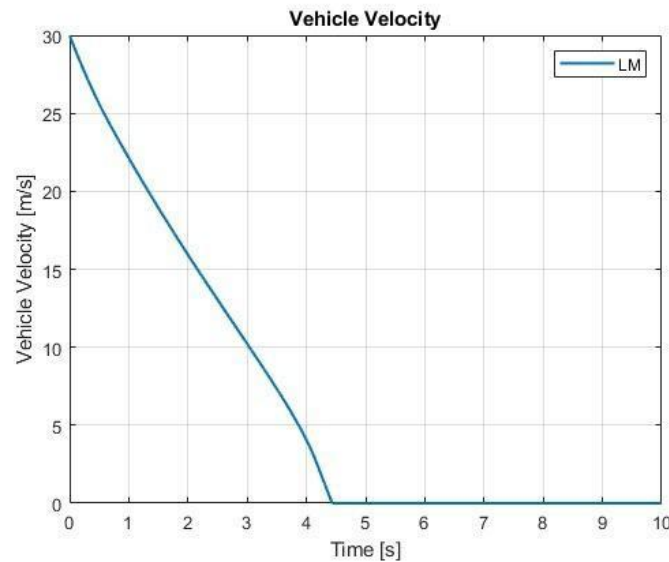
```
52 % Vehicle longitudinal dynamics
53 Fbr_total = sum(Fxf_vec);
54 a = -Fbr_total / m;
55
56 Velocity = Velocity_prev + a * dt;
57 Velocity = max(min(Velocity, Vel_max), Vel_min);
58
59 % Updated states
60 Wheel_L_prev = Wheel_L;
61 Wheel_R_prev = Wheel_R;
62 Velocity_prev = Velocity;
63
64 end
```

Fig: This block represents the vehicle dynamics used for study

Results and Conclusions

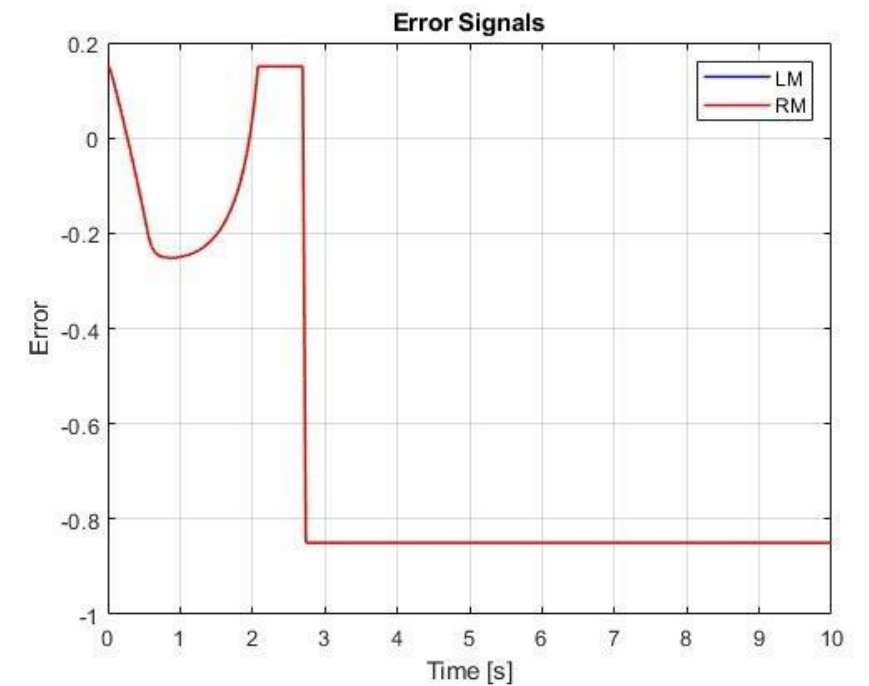
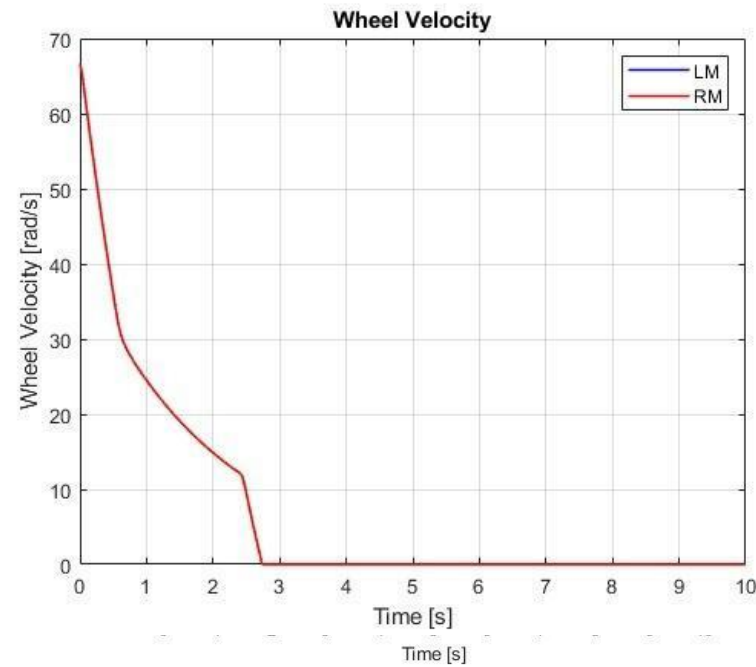
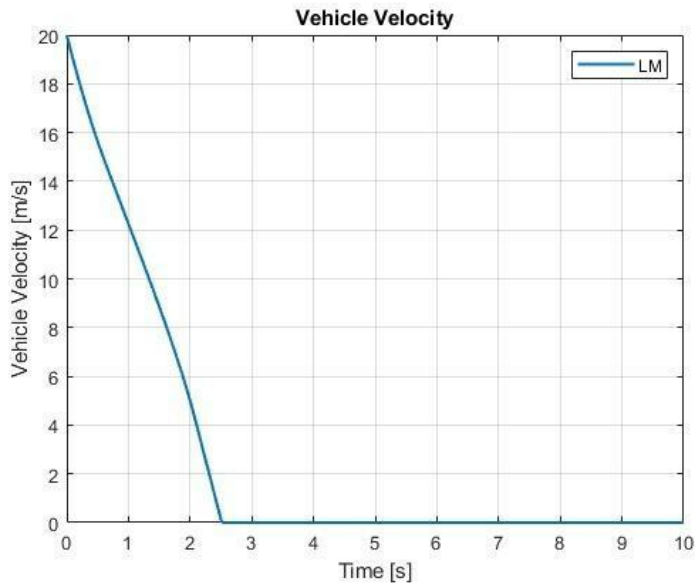
Performance Under Parameter Variations

Effect of Initial Velocity



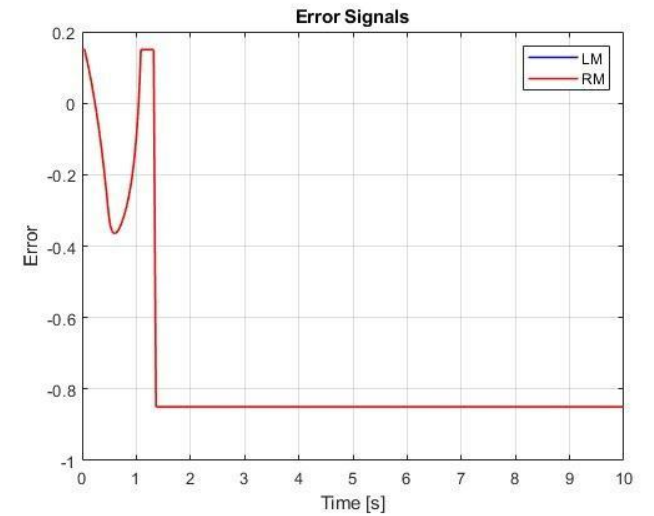
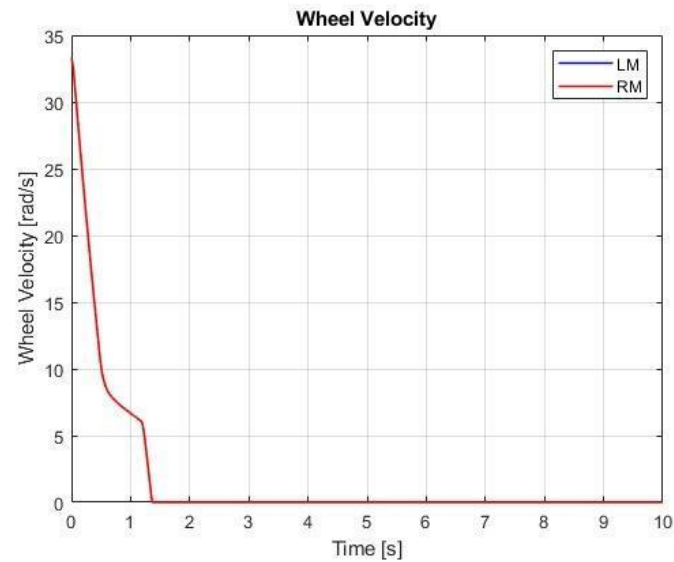
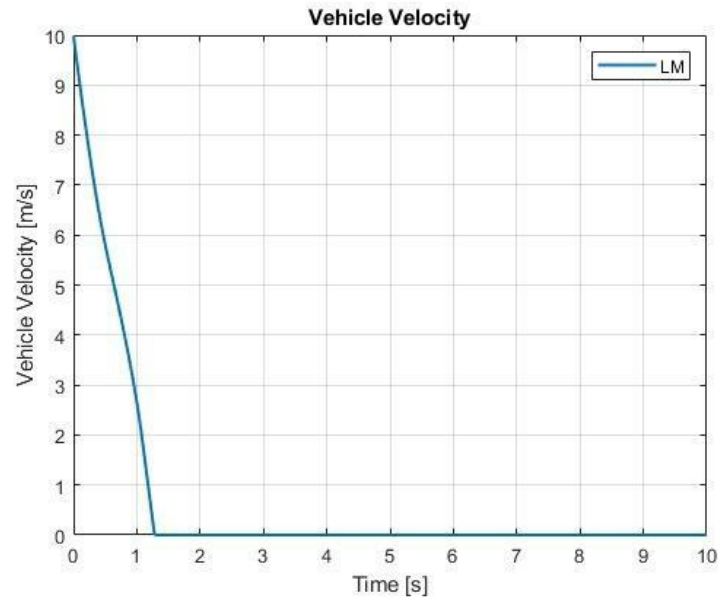
Case 1: When initial velocity is kept Higher (30 m/s)
The time taken by the vehicle reaches to 4.3 sec.

Effect of Initial Velocity



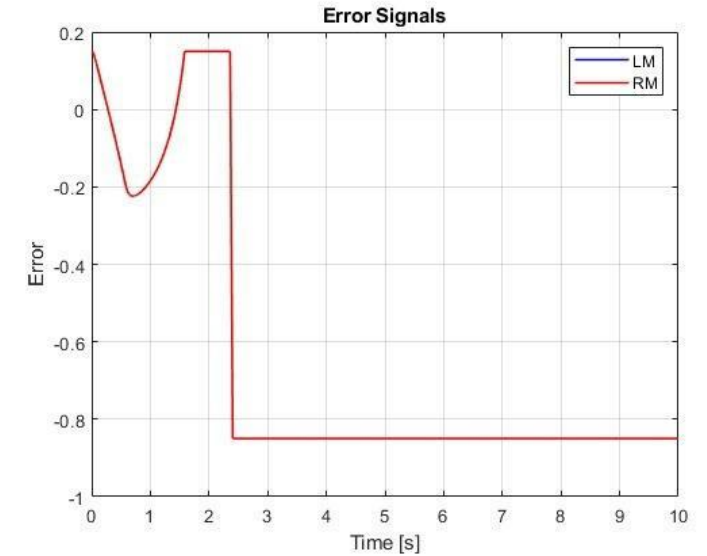
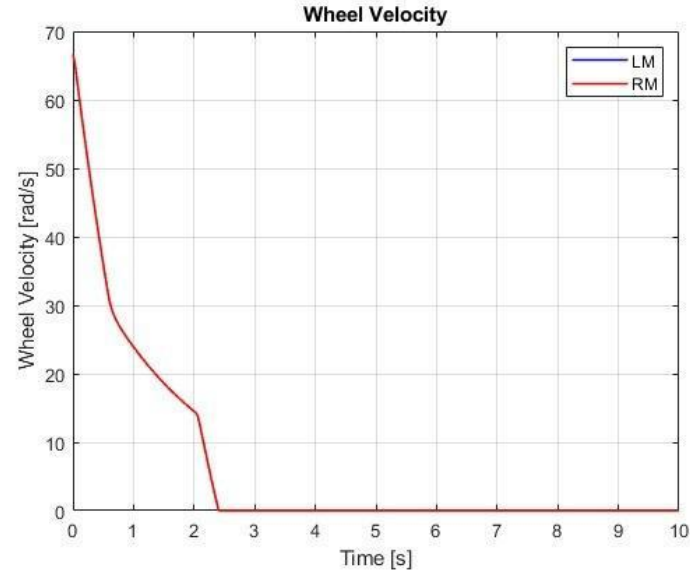
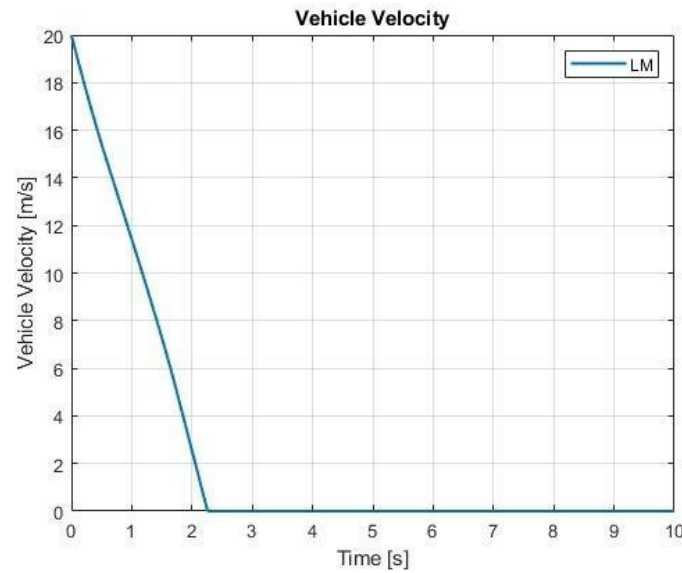
Case 2: When initial velocity is kept Nominal (20 m/s)
The time taken by the vehicle reaches to 2.4 sec.

Effect of Initial Velocity



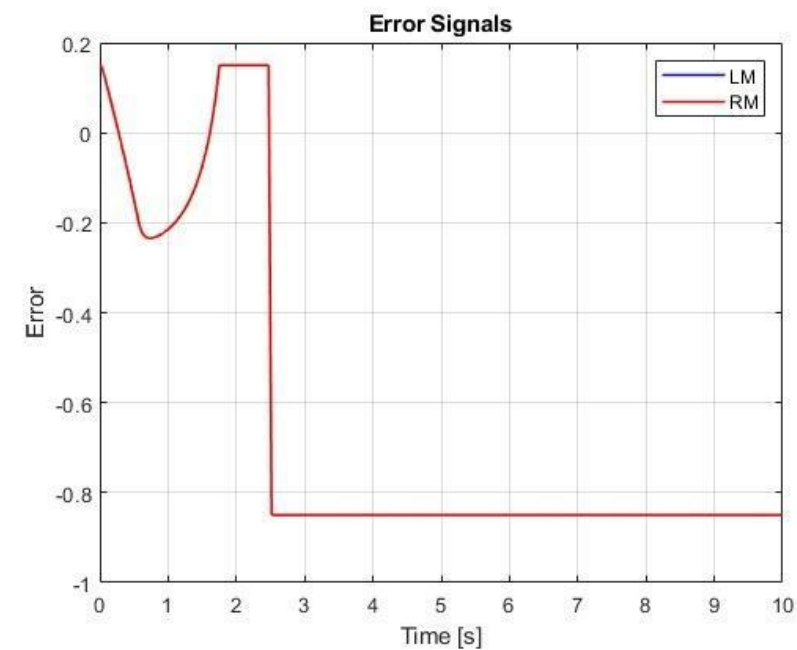
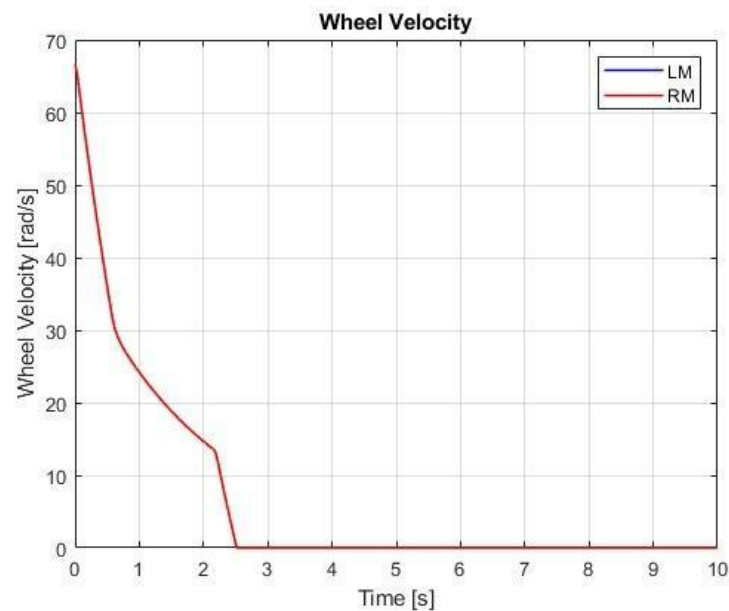
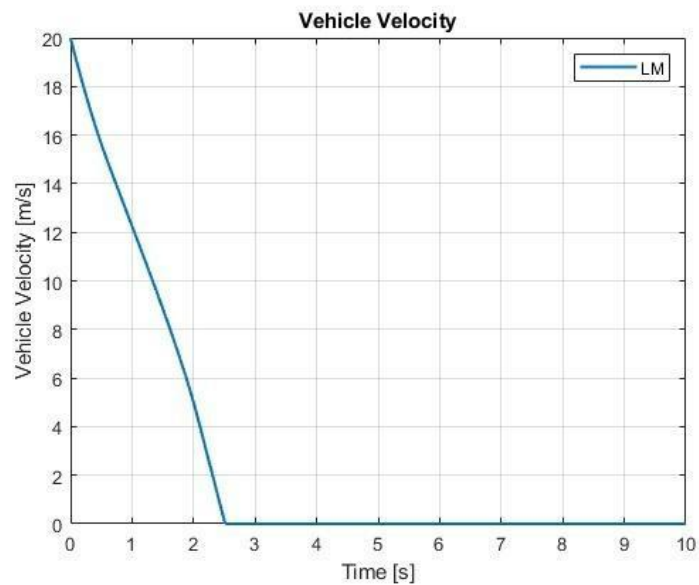
Case 3: When initial velocity is kept Low (10 m/s)
The time taken by the vehicle reaches to 1.1 sec.

Effect of Changing Friction Coefficient



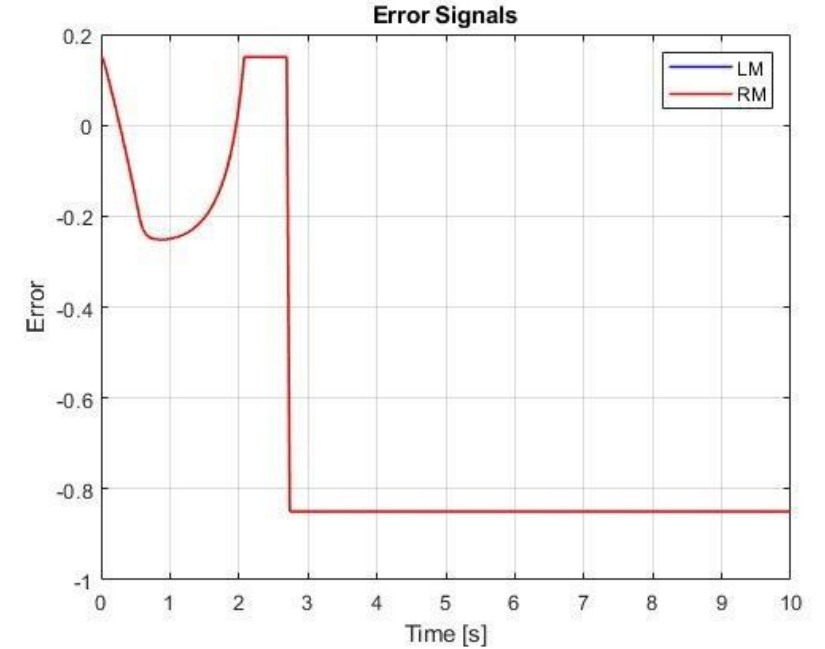
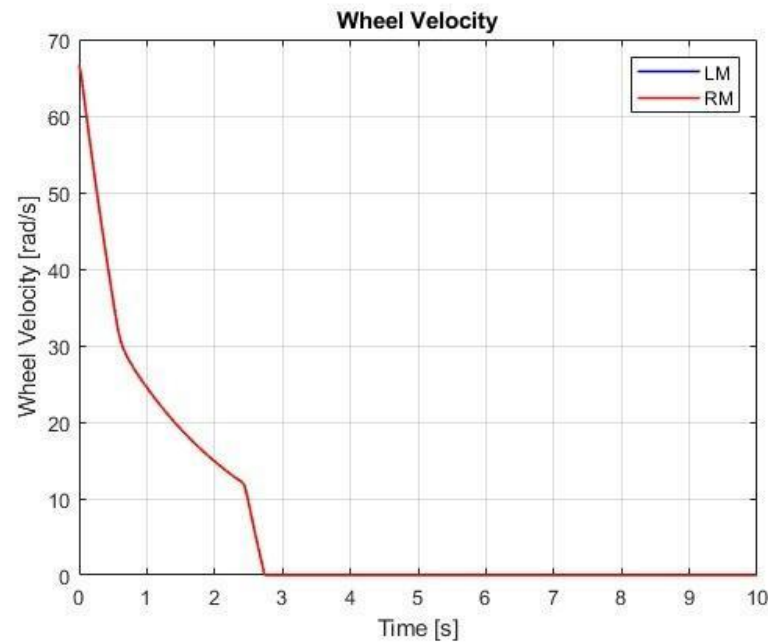
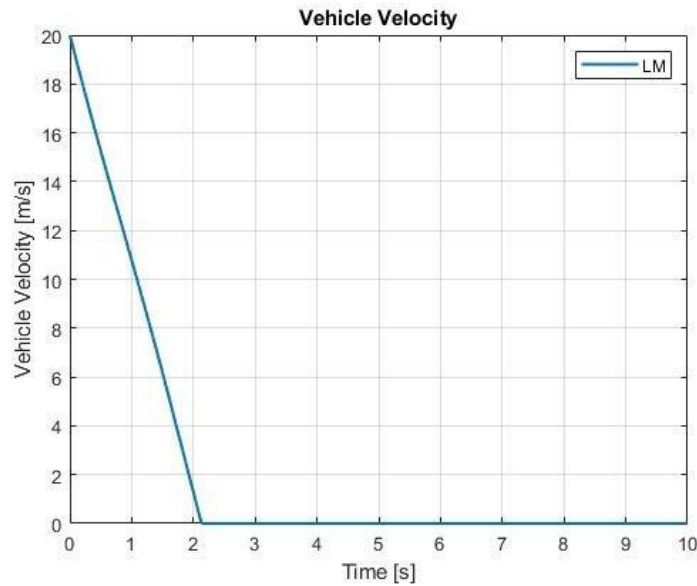
Case 1: When runway is dry ($\mu = 0.8$)
The time taken by the vehicle reaches to 2.2 sec.

Effect of Changing Friction Coefficient



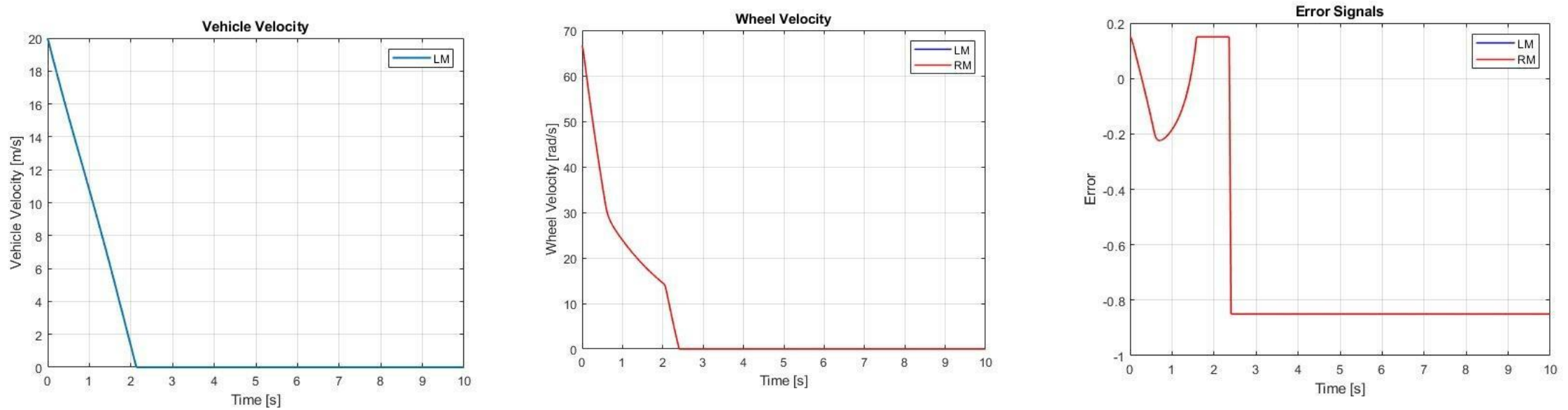
Case 2: When runway is wet ($\mu = 0.5$)
The time taken by the vehicle reaches to 2.4 sec.

Effect of Changing Friction Coefficient



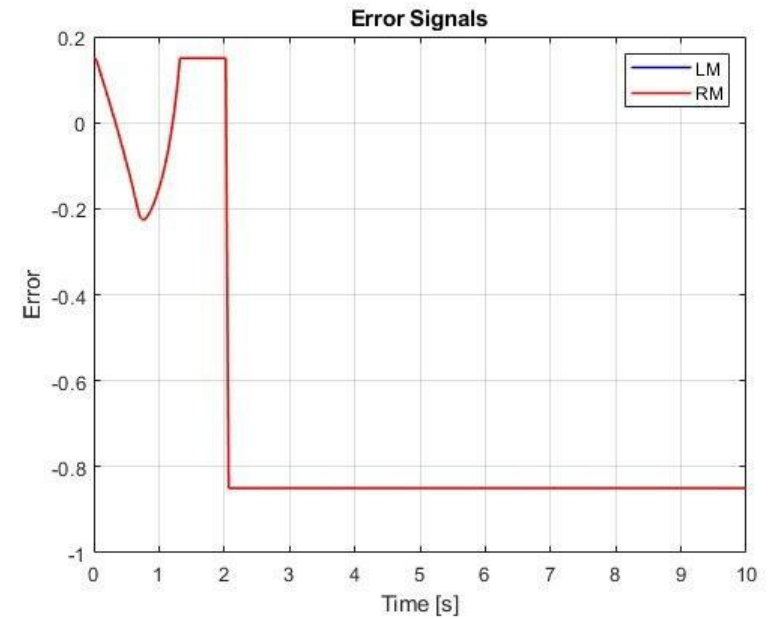
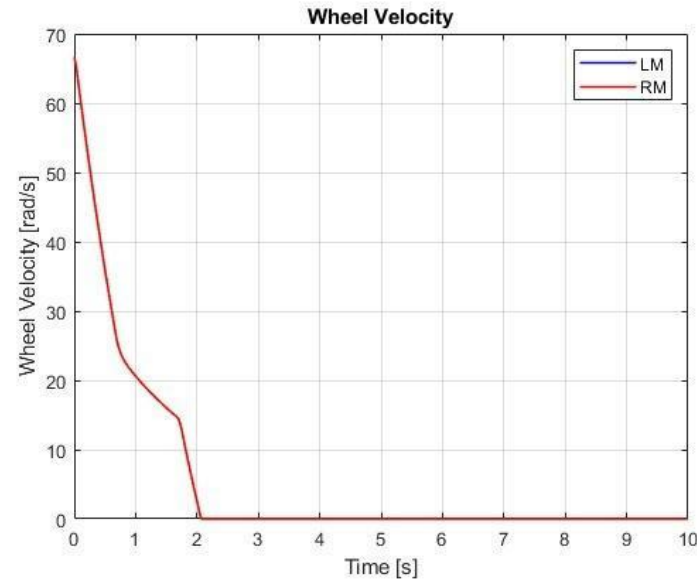
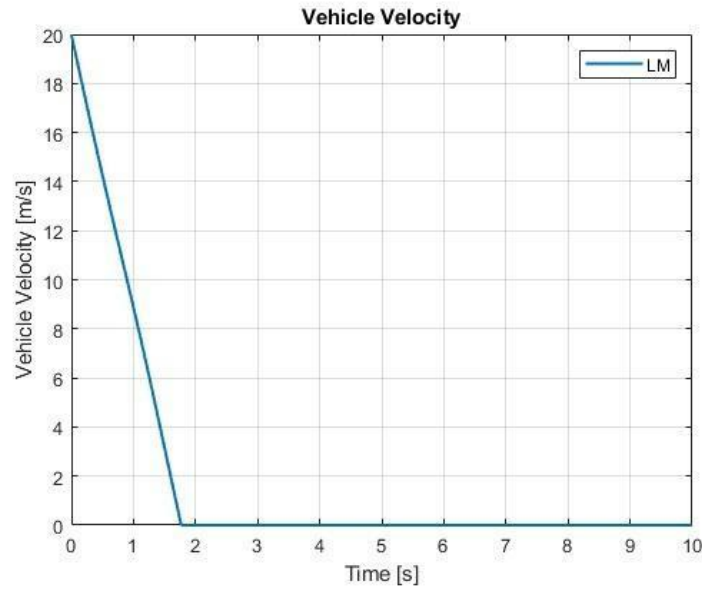
Case 3: When runway is sandy ($\mu = 0.2$)
The time taken by the vehicle reaches to 2.8 sec.

Effect of Wheel Load Variations



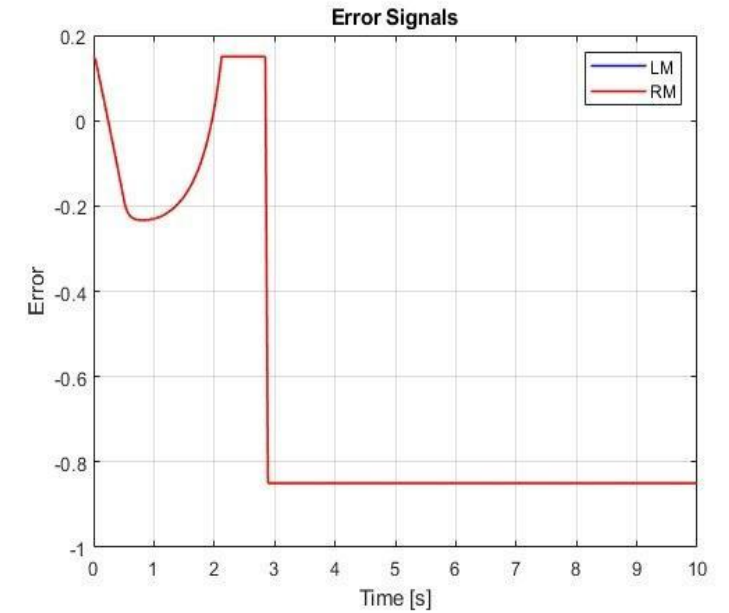
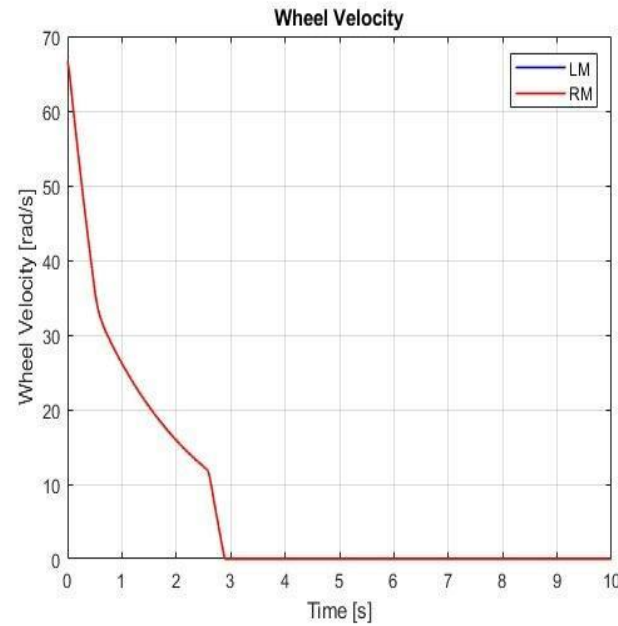
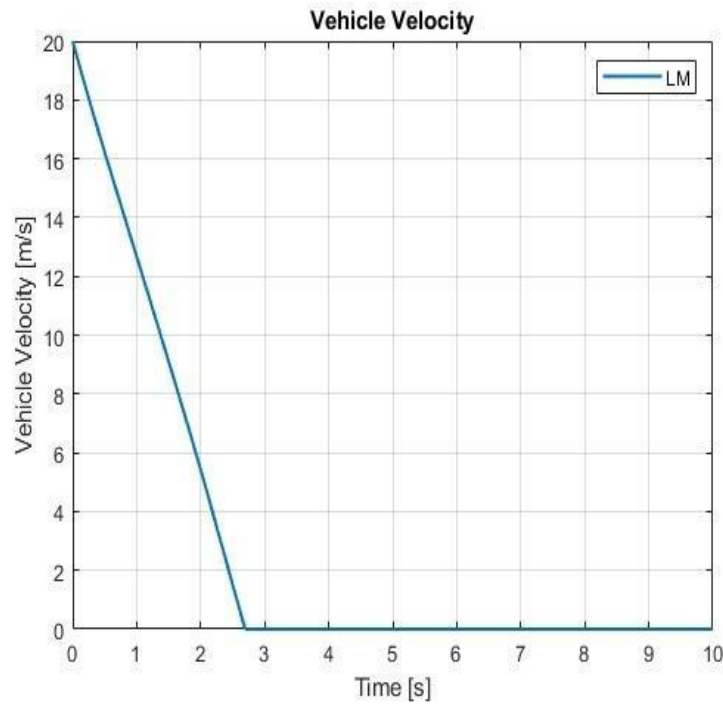
Case 1: When load on wheel is kept high (9000N)
The time taken by the vehicle reaches to 1.8 sec.

Effect of Wheel Load Variations



Case 2: When load on wheel is kept nominal (7500N)
The time taken by the vehicle reaches to 2.2 sec.

Effect of Wheel Load Variations

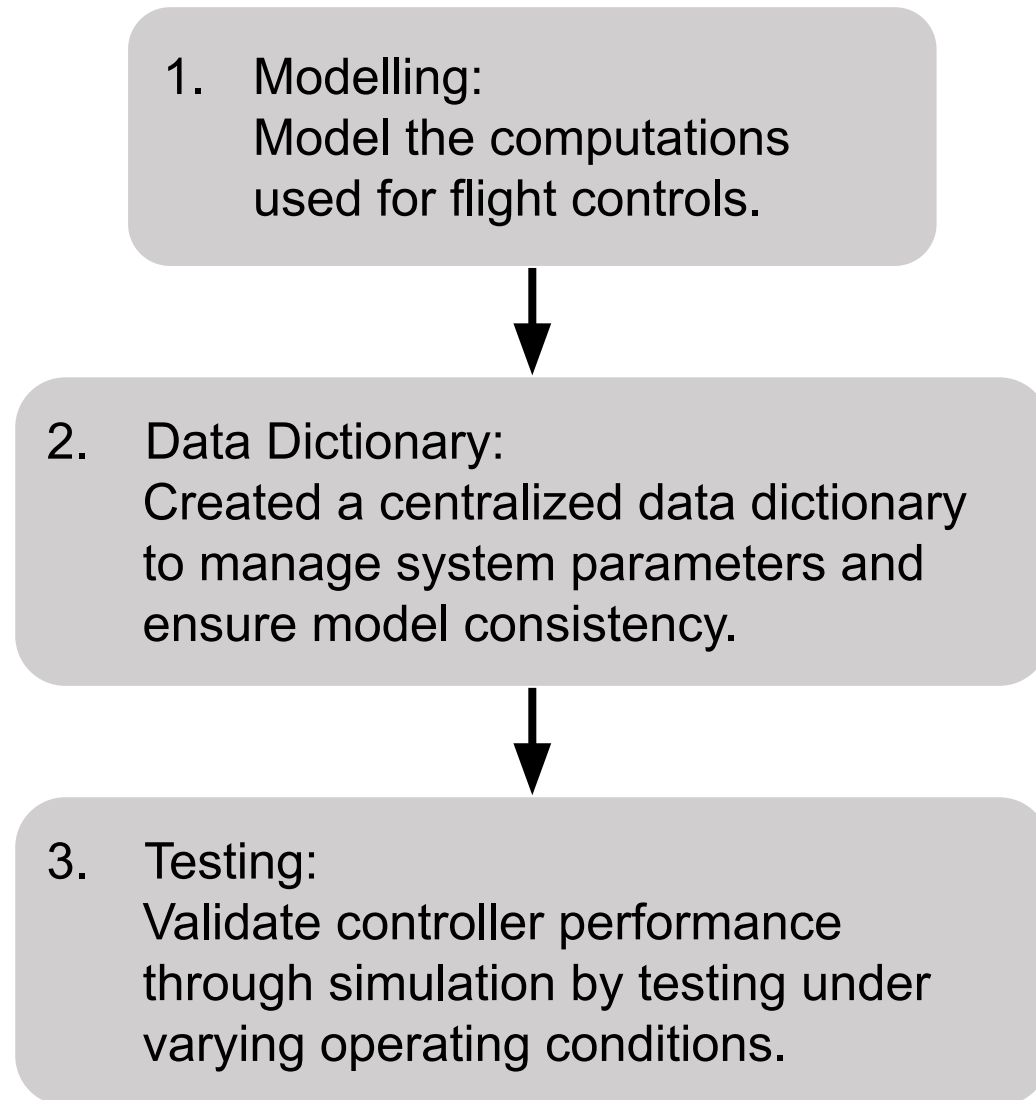


Case 3: When load on wheel is kept low (6000N)
The time taken by the vehicle reaches to 2.7 sec.

OBSERVATION & KEY FINDINGS

Parameter	Variation	Stopping Time(s)
1) Effect of initial velocity	High Speed	4.3
	Nominal Speed	2.4
	Low Speed	1.1
2) Effect of Friction Coefficient	Dry Runway	2.2
	Wet Runway	2.4
	Sandy Runway	2.8
3) Effect of Wheel Load	Higher Wheel Load	1.8
	Nominal Wheel Load	2.2
	Lower Wheel Load	2.7

Departmental Projects



Objectives

$$M = \frac{K1\delta_e + K2\delta_a + K3V + K4\alpha + K8\ln(1+V) + K9\alpha}{K5 - (K6\delta_e + K7\delta_a)}$$

K1: Elevator control
K2: Aileron cross-coupling effect
K3: Dynamic pressure
K4: Pitching moment per angle of attack
K5: Base stability margin
K6: Elevator damping effect
K7: Aileron damping effect
K8: Nonlinear dynamic pressure
K9: Nonlinear angle-of-attack

δ_e : Elevator deflection (rad)
 δ_a : Aileron deflection (rad)
V: Airspeed (m/s)
 α : Angle of attack (rad)

Fig: Flight Control Computation

Modelled Computation

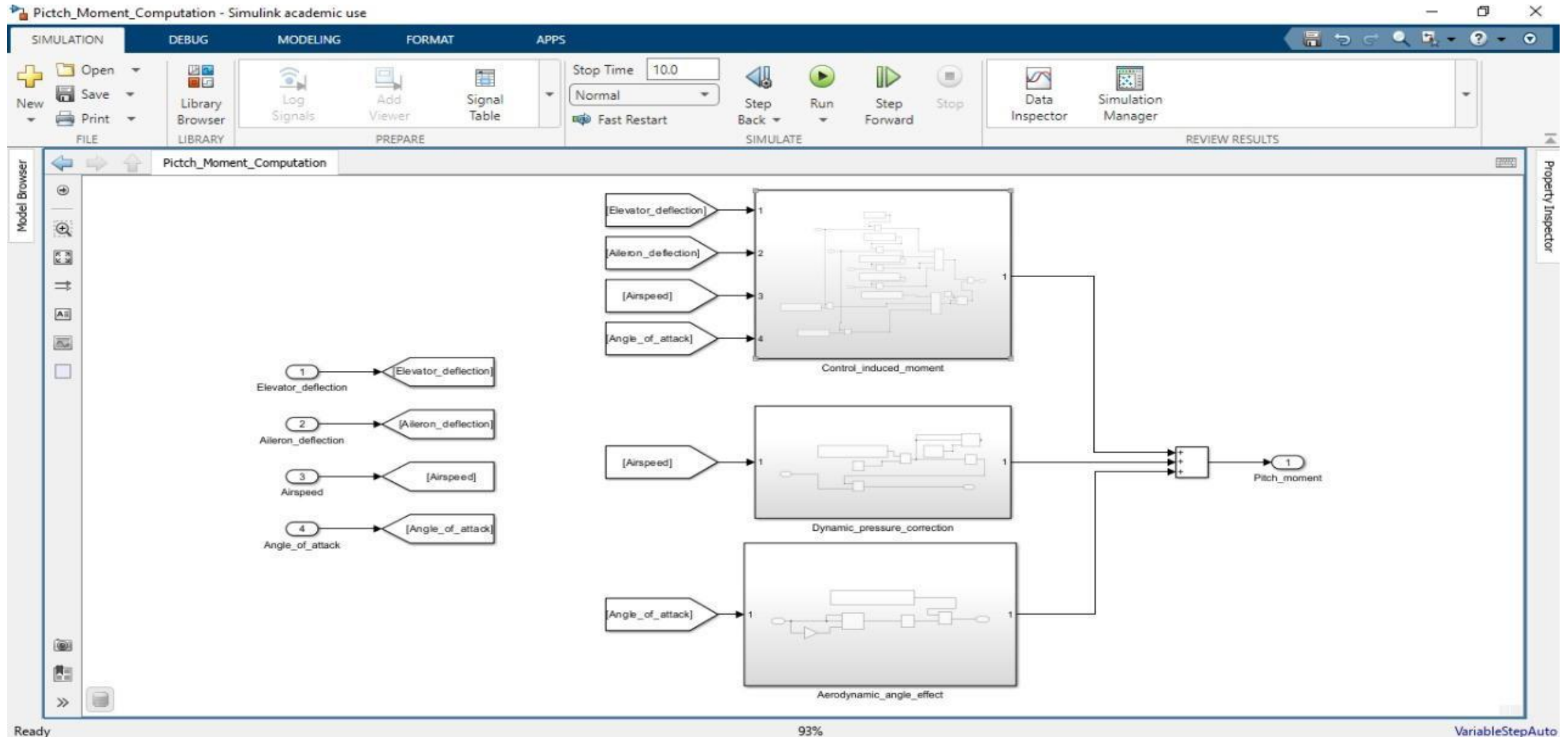


Fig: Modelled computations on Simulink

Data Dictionary

Model Explorer

File Edit View Tools Add Help

Model Hierarchy

- Simulink Root
 - Base Workspace
 - Pitch_moment_data_dictionary
 - Design Data
 - Architectural Data
 - Brake_model_UCAV
 - Pitch_Moment_Computer
 - Configurations
 - Model Workspace
 - External Data
 - Base Workspace
 - Pitch_moment_data_dictionary
 - Aerodynamic_angle_of_attack
 - Control_induced_moment
 - Dynamic_pressure_coefficient

Contents of: ...esign 'C:\Users\dell\Desktop\Pitch_moment_data_dictionary.sldd' (only) Filter Contents

Column View: Dictionary Objects Show Details 9 object(s)

Name	Status	Value	DataType	Dimensions	Complexity	Min	Max	Units
Elevator_control	New	0.0034	auto	[1 1]	real	[]	[]	
Dynamic_pressure	New	0.0056	auto	[1 1]	real	[]	[]	
Pitching_moment_per_angle_of_attack	New	0.0079	auto	[1 1]	real	[]	[]	
Base_stability_margin	New	0.0067	auto	[1 1]	real	[]	[]	
Elevator_damping_effect	New	0.0052	auto	[1 1]	real	[]	[]	
Aileron_damping_effect	New	0.0039	auto	[1 1]	real	[]	[]	
Aileron_cross_coupling_effect	New	0.0012	auto	[1 1]	real	[]	[]	
Non_linear_dynamic_pressure_factor	New	0.0046	auto	[1 1]	real	[]	[]	
Non_linear_angle_of_attack_factor	New	0.0075	auto	[1 1]	real	[]	[]	

This section of the dictionary is visible to the user. This section of the dictionary is visible to the user.

Revert

Contents Search Results

Fig: Data dictionaries for all the constants and variable used for computation

Results

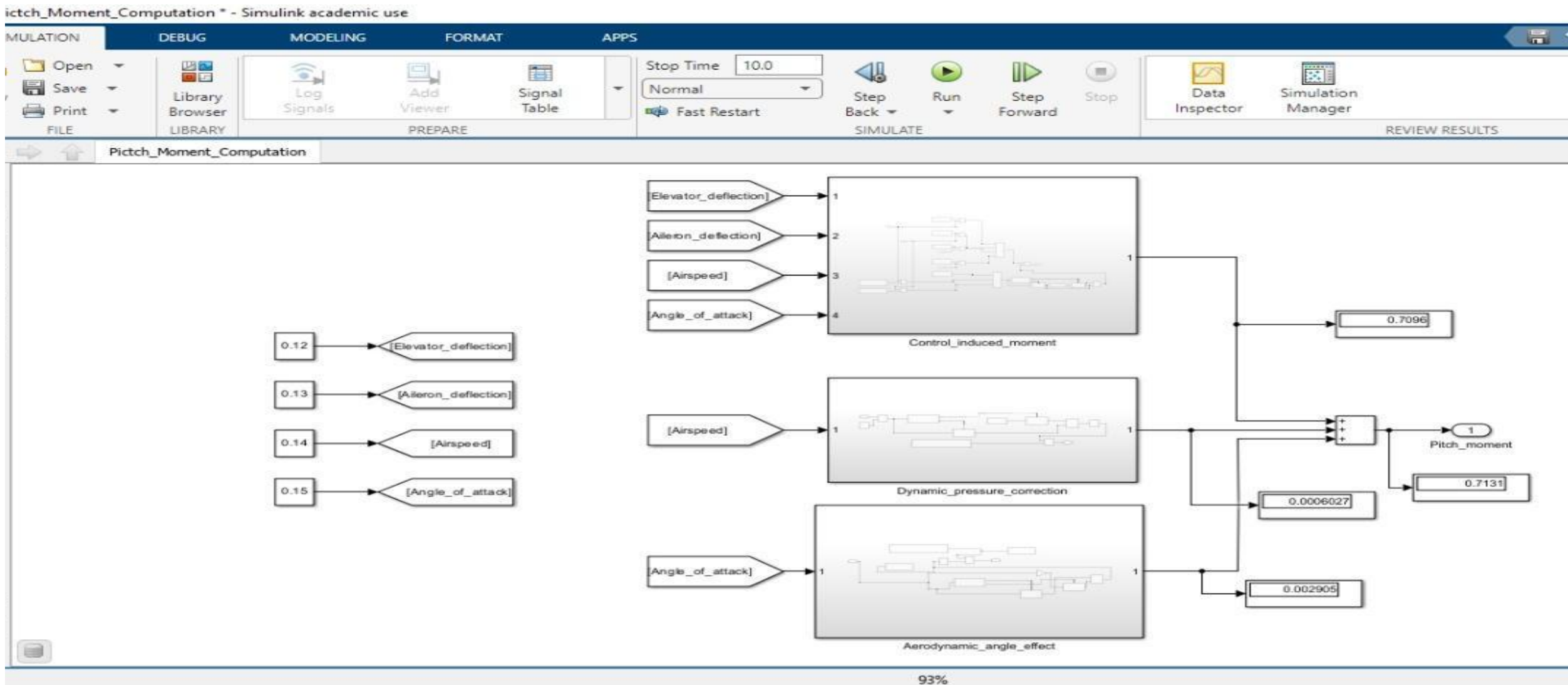


Fig: Result after computation using Simulink

Validation

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW ADD-INS						
Paste Cut Copy Format Painter		Calibri 11 A^A B I U				General %
Clipboard		Font		Alignment		Number
F4		✕ ✓ f _x		=E3+E4+E5		
	A	B	C	D	E	F
1						
2	Elevator_deflection	0.12				
3	Aileron_deflection	0.13			0.709572	
4	Airspeed	0.14			0.000603	0.713079146
5	Angle_of_attack	0.15			0.002905	
6						
7	Elevator_control	0.0034				
8	Dynamc_pressure	0.0056				
9	Pitching_moment_per_angle_of_attack	0.0079				
10	Base_stabiity_margin	0.0067				
11	Elevtor_damping_effect	0.0052				
12	Aileron_damping_effect	0.0039				
13	Aileron_cross_coupling_effect	0.0012				
14	Non-linear_dynamic_pressure_factor	0.0046				
15	Non-linear_angle_of_attack_factor	0.0075				

Fig: Results validated using Ms excel

Learnings From the Internship

- Learned model-based system development using MATLAB/Simulink
- Developed skills in designing and tuning PID control laws to achieve stable and responsive closed-loop system behavior.
- Gained practical insight into closed-loop wheel brake systems, including braking dynamics, feedback control, and stability considerations.
- Understood the integration of actuators, sensors, and control algorithms within a closed-loop control framework.

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- Jung Jin Lee, Jung-hoon Lee, “Design Study of Brake Mechanical Control System of UAV”, 33rd Congress of the International Council of Aeronautical Science Stockholm, Sweden, 2022

Thank You